Analysing Water Potability to Support SDG 6 Clean Water and Sanitation & SDG 3 Good Health and Wellbeing

HW 2 Group 1

1. Model Selection & Justification

- **Random Forest:** It is an ensemble method that builds multiple decision trees and combines their outputs. It handles non-linear relationships between features well and is robust to outliers. These are important qualities for environmental data where relationships between pH, sulfate, and potability are complex and non-linear.

- **Logistic Regression:** Logistic Regression serves as a baseline model. It's simple, interpretable, and fast to train. Comparing it against random forest helps demonstrate whether the problem requires a more complex model or whether a linear boundary is sufficient.

The goal is to correctly identify unsafe water sources so that public health interventions can be prioritised in water stressed regions

2. Data Preparation

• An 80/20 train-test split was applied.2,620 training, 656 test samples

• All features were standardised using StandardScaler to ensure no single feature dominates due to scale differences.

• To contextualize the data within a global framework, we merged the potability data with the World Bank Water Stress dataset. The data measures how much of a country's available freshwater is consumed. However, the original potability dataset contains no geographic information. There are no country names so this made a geographically accurate merge impossible. The only available approach was to randomly assign country codes using np.random.seed(42). This was not a preferred choice but the merge was still performed to practise combining datasets in a real pipeline.

3. Model Performance Results

Model	**Accuracy**	**Precision (Potable)**	**Recall (Potable)**	**F1 (Potable)**
Random Forest	68.4%	0.66	0.32	0.43
Logistic Regression	62.8%	0.00	0.00	0.00

Random Forest

Random forest achieved a ROC AUC of 0.68 (68.4% overall accuracy) correctly identifying 90% of non potable samples (recall = 0.90). This is important because it prevents people from drinking contaminated water.

However, recall for Potable water was only 0.32 (this means the model missed 68% of actually safe samples, classifying them as unsafe) In a public health context, this may be acceptable since it is safer to flag unsafe water rather than to miss it

Logistic Regression

Achieved 62.8% accuracy but failed to predict any safe water samples, classifying everything as Not Potable

Our ROC Curve confirmed Random forest (AUC = 0.68) significantly outperformed Logistic Regression (AUC = 0.52), which performed slightly better than random guessing

4. Interpretation & SDG Insights

The results prove that water safety is a complex, non-linear problem. Random forest is the superior choice for identifying safe drinking sources.

Insights: We found that water quality and water stress must be measured together. The weak linear correlations (max 0.03) found in our heatmap confirms that water safety depends on multiple interactions rather than single parameter.

High water stress (like the UAE at 1587%) means that any "Not Potable" classification further cripples a nation's usable supply. In such high stress regions, AI can optimize resource allocation by instantly flagging non-potable water via basic sensor readings (conductivity, turbidity) instead of slow, expensive lab tests.

Limitations: The model's low recall for potable water was restricted by the 60/40 class imbalance. Future improvements could include SMOTE or balanced class weights to improve detection of safe water.

5. Conclusion

While logistic regression shows a 62.8% accuracy, the confusion matrix reveals a true positive count of zero, meaning it failed to find any safe water sources.

In contrast, the random forest model successfully identified 77 potable samples**,** proving its ability to parse the non-linear chemical interactions that determine water safety.

This distinction is vital for SDG's. we need models that can actually detect safe water, not just guess the most common outcome.

Ultimately, **random forest** is the recommended architecture. It demonstrates that advanced AI models are required to ensure that in water-stressed regions, the remaining water is truly safe to consume.